THE TRANSCENDENCE-DEGREE DIMENSION FUNCTION ON A DECENT SPACE OVER A UNIVERSALLY CATENARY BASE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7365-7457, immediately following the statement at 7348-7363.

Editorial note. The statement contains the author's typo 'transcendence degreeof x/f(x)' (missing space); it is retained verbatim.

Editorial note. Ten cross-chapter pointers are recorded in assembly.external_references; all are navigation pointers and the chain-lifting argument itself is inside the span.

Allowed standard prerequisites (editorial).

- Dimension functions and their additivity along fibres. Morphisms of Spaces, definition spaces-morphisms-definition-dimension-fibre and lemma spaces-morphisms-lemma-dimension-fibre-at-a-point-additive, cited in the statement and proof. Defines delta_X as delta plus transcendence degree and supplies additivity along specialisations inside fibres.
- Catenarity of the base transfers to the total space. Decent Spaces lemmas decent-spaces-lemma-locally-Noetherian-decent-quasi-separated and decent-spaces-lemma-check-dimension-function-finite-cover, cited inside the proof. Reduces catenarity questions to a finite cover and lets the argument work on decent spaces rather than schemes.
- Existence of a scheme finite over the space. Limits of Spaces, proposition spaces-limits-proposition-there-is-a-scheme-finite-over, cited inside the proof. Produces the finite morphism pi' : Y' -> X used to lift the two chains of specialisations to a common object.
- Constructing the gluing used in the reduction. chapter lemma lemma-construct-glueing, cited inside the proof. Builds the finite Y' with prescribed behaviour away from the fibre over the chosen point.

Source reference. Original reference 0ED9: focused-seeds/geometry/S9-spaces-univ-catenary-dim-func/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ACK: focused-seeds/geometry/S9-spaces-univ-catenary-dim-func/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AHB: focused-seeds/geometry/S9-spaces-univ-catenary-dim-func/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the

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counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BB6: focused-seeds/geometry/S9-spaces-univ-catenary-dim-func/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ED5: focused-seeds/geometry/S9-spaces-univ-catenary-dim-func/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0EDM: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7274–7274. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02IL: focused-seeds/geometry/S9-spaces-univ-catenary-dim-func/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09YC: focused-seeds/geometry/S9-spaces-univ-catenary-dim-func/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NM: focused-seeds/geometry/S9-spaces-univ-catenary-dim-func/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NR: focused-seeds/geometry/S9-spaces-univ-catenary-dim-func/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NZ: focused-seeds/geometry/S9-spaces-univ-catenary-dim-func/ref-tags.txt, source lines 13–13. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let $\delta : |B| \to \mathbf{Z}$ be a function. Assume B is decent, locally Noetherian, and universally catenary and δ is a dimension function. If X is a decent algebraic space over B whose structure morphism $f : X \to B$ is locally of finite type we define $\delta_X : |X| \to \mathbf{Z}$ by the rule*

$$\delta_X(x) = \delta(f(x)) + \textit{transcendence degree}_{\textit{of } x/f(x)}$$

(Morphisms of Spaces, Definition 04NM). Then δ_X is a dimension function.

Proof. The problem is local on B. Thus we may assume B is quasi-compact. By Decent Spaces, Lemma 0BB6 we see B is quasi-separated. By Limits of Spaces, Proposition 09YC we can choose a finite surjective morphism $\pi : Y \to X$ where Y is a scheme. Claim: δ_Y is a dimension function.

The claim implies the lemma. With $X \to B$ as in the lemma set $Z = Y \times_B X$ with projections $p : Z \to Y$ and $q : Z \to X$. Then we have

$$\delta_Z(z) = \delta_Y(p(z)) + \text{transcendence degreeof } z/p(z)$$

and $\delta_Z(z) = \delta_X(q(z))$. This follows from Morphisms of Spaces, Lemma 04NR and the fact that these transcendence degrees are zero for finite morphisms. By Decent Spaces, Lemma 0ED5 and the claim we find that δ_Z is a dimension function. Then we find that δ_X is a dimension function by Decent Spaces, Lemma 0ED9.

Proof of the claim. Consider a specialization $y \rightsquigarrow y'$, $y \neq y'$ of points of the Noetherian scheme Y. Then $\delta_Y(y) > \delta_Y(y')$ because there are no specializations between points in fibres of Y (see Decent Spaces, Lemma 0ACK). Using this for a chain of specializations we find

$$\delta_Y(y) - \delta_Y(y') \geq \text{codim}(\overline{\{y'\}}, \overline{\{y\}})$$

Our task is to show equality. By Properties, Lemma 02IL we can choose a specialization $y' \rightsquigarrow y_0$. It suffices to show $\delta_Y(y) - \delta_Y(y_0) = \text{codim}(\overline{\{y_0\}}, \overline{\{y\}})$ because this will imply the equality for both $y \rightsquigarrow y'$ and $y' \rightsquigarrow y_0$.

Choose a maximal chain $y = y_c \rightsquigarrow y_{c-1} \rightsquigarrow \ldots \rightsquigarrow y_0$ of specializations in Y. Set $b = \pi(y)$ and $b_0 = \pi(y_0)$. Choose a maximal chain $b = b_e \rightsquigarrow b_{e-1} \rightsquigarrow \ldots \rightsquigarrow b_0$ of specializations in $|B|$. We have to show $e = c$. Since π is closed (Morphisms of Spaces, Lemma 04NZ) we can find a sequence of specializations $y = y'_e \rightsquigarrow y'_{e-1} \rightsquigarrow \ldots \rightsquigarrow y'_0$ mapping to $b = b_e \rightsquigarrow b_{e-1} \rightsquigarrow \ldots \rightsquigarrow b_0$. Observe that $y'_e \rightsquigarrow y'_{e-1} \rightsquigarrow \ldots \rightsquigarrow y'_0$ is a maximal chain as well. If $y_0 = y'_0$, then because Y is catenary, we conclude that $e = c$ as desired. In the next paragraph we reduce to this case by sleight of hand and we conclude in the same manner.

Since π is closed we see that b_0 is a closed point of $|B|$. By Decent Spaces, Lemma 0AHB we can represent b_0 by a closed immersion $b_0 : \text{Spec}(k) \to B$. By Lemma 0EDM we can find a factorization

$$Y \to Y' \to X$$

with $\pi' : Y' \to X$ finite and $Y \to Y'$ a morphism which map y_0 and y'_0 to the same point and is an isomorphism away from the inverse image of b_0. (Of course Y' won't be a scheme but this doesn't matter for the argument that follows.) Clearly the maximal chains of specializations $y_c \rightsquigarrow y_{c-1} \rightsquigarrow \ldots \rightsquigarrow y_0$ and $y'_e \rightsquigarrow y'_{e-1} \rightsquigarrow \ldots \rightsquigarrow y'_0$ map to maximal chains of specializations in Y' having the same start and end. Since B is universally catenary, we see that $|Y'|$ is catenary and we conclude as before. $\square$